

CAPSTONE PROJECT / ENTERPRISE AI RAG SUPPORT

SCHOOLTRUST AI

Verified Parent Support and Knowledge Improvement System

A controlled setup sequence for the assessed prototype.

Project	SchoolTrust AI
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Target platform	n8n + Supabase + OpenAI + static web interface
Version / date	Version 1.0 14 July 2026
Estimated setup	45 to 90 minutes after credentials are available

SETUP AND DEPLOYMENT GUIDE

One focused project. Approved sources. Human-controlled improvement.

1. Package Contents

Path	Purpose
n8n/SchoolTrust_AI_Verified_Parent_Support.json	Importable automation with parent, administrator, dashboard and regression branches.
database/supabase_schema.sql	Tables, vector function, dashboard function, gap grouping and test seed.
web/	Responsive parent and administrator prototype.
data/	Draft starter knowledge and regression-test reference CSV files.
docs/	SRS, setup guide, demo script and checklist.

2. Prerequisites

- An n8n instance that permits webhook, schedule, code and HTTP Request nodes.
- A Supabase project with permission to enable pgvector.
- An OpenAI API key with embedding and chat-model access.
- A static hosting location or access to the existing school website.
- Administrator-approved school information. Do not publish unreviewed draft data.

3. Configure Supabase

- Open the Supabase SQL editor.
- Review database/supabase_schema.sql. Use a new project when possible.
- Execute the script once.
- Confirm that six tables exist: knowledge_chunks, conversation_logs, knowledge_gaps, regression_tests, evaluation_results and admin_actions.
- Confirm that match_school_knowledge, record_knowledge_gap and schooltrust_admin_dashboard functions exist.
- Keep the service-role key server-side. Never place it in the web folder.

Important

The included starter knowledge CSV is marked draft. Publish each reviewed entry through the administrator approval endpoint so its embedding and approval metadata are created.

4. Import and Configure n8n

- Import n8n/SchoolTrust_AI_Verified_Parent_Support.json.
- Keep the workflow inactive during configuration.
- Set SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, OPENAI_API_KEY, OPENAI_CHAT_MODEL, OPENAI_EMBEDDING_MODEL and SCHOOLTRUST_ADMIN_TOKEN.
- If custom environment values are unavailable, replace the expressions with encrypted n8n credentials. Do not hardcode secrets in the browser.
- Execute each HTTP node once during testing and check for a 2xx response.
- Run the Manual Quality Test trigger after approved knowledge exists.
- Activate the workflow only after the ask, dashboard and approval production webhooks respond correctly.

Variable	Example / purpose
SUPABASE_URL	https://YOUR_PROJECT.supabase.co

Variable	Example / purpose
SUPABASE_SERVICE_ROLE_KEY	Server-side database access. Treat as secret.
OPENAI_API_KEY	Server-side AI access. Treat as secret.
OPENAI_CHAT_MODEL	gpt-4.1-mini or another JSON-capable approved model.
OPENAI_EMBEDDING_MODEL	text-embedding-3-small, matching vector(1536).
SCHOOLTRUST_ADMIN_TOKEN	A long random token used only for the prototype administrator endpoints.

5. Publish Initial Knowledge

Use the Administrator Approval Webhook. Required body fields:

Field	Example
document_title	Annual Fund Policy
category	fees
content	The exact administrator-approved answer.
actor	School Administrator
effective_from	2026-01-01
expires_at	Optional
gap_id	Optional UUID when resolving a gap

Recommended first records

School profile, official contact, classes offered, annual-fund categories and one active admissions policy. Keep time-sensitive notices in separate entries with expiry dates.

6. Configure the Web Interface

- Open web/config.js.
- Replace https://YOUR-N8N-DOMAIN with the n8n production base URL.
- Replace the demo administrator token with the same SCHOOLTRUST_ADMIN_TOKEN value.
- Deploy the web folder to a static host or the existing website.
- For production, replace the browser-held administrator token with a real authenticated administrator session.

7. Required Test Sequence

Test	Expected result
Ask the annual-fund question	Both categories are stated or a precise clarification is requested.
Ask an unsupported transport-fee question	Safe escalation plus an open knowledge gap.
Load dashboard without token	Request rejected.
Load dashboard with valid token	Live metrics, gaps and test results returned.

Test	Expected result
Approve the transport answer	Approved knowledge stored, embedding created and related gap resolved.
Run manual quality test	One evaluation result stored for each active test.
Ask the transport question again	Verified answer returned if the new source meets retrieval and confidence rules.

8. Troubleshooting

Problem	Check
Web interface says setup required	Confirm web/config.js contains the production n8n base URL.
401 or unauthorized administrator request	Confirm x-admin-token matches SCHOOLTRUST_ADMIN_TOKEN.
Supabase RPC not found	Run the complete SQL schema and refresh the Supabase API cache.
No knowledge retrieved	Confirm status is approved, dates are active and embedding is not null.
Vector dimension error	Use text-embedding-3-small with vector(1536), or update both schema and model together.
Model response cannot be parsed	Use a chat model supporting JSON-object response format.
Quality tests all fail	Publish approved source records before running the test branch.